```
D:\Software\Python\Python311\python.exe E:\PythonProjectsByYear\Year2025\MC-PIKAN_Official\experiments\Volterra1D_MCPI.py
The network model is PINNs
Layers: ModuleList(
  (0): Linear(in_features=1, out_features=20, bias=True)
  (1): Linear(in_features=20, out_features=20, bias=True)
  (2): Linear(in_features=20, out_features=1, bias=True)
)
Activation function: Tanh()
Total trainable parameters: 481
Model on device: cuda:0
Start training...
Max Epochs: 2000, Learning Rate: 0.001, Epsilon: 1e-20, Optimizer: Adam
Training: 100%|████████████████████| 2000/2000 [00:06<00:00, 323.10epoch/s, Loss=1.28e-04, LR=1.0e-03, Best=1.28e-04]
Training finished. Best loss: 8.95e-05. Model saved to results/best_model.pth
Loss history saved to results/loss_history.npy
Training time: 7.052142143249512s
Relative loss: 0.01178472675383091
L2 loss: 0.007498638238757849
Parameter containing:
tensor([[-1.2512],
        [-0.1482],
        [ 0.8240],
        [-1.1389],
        [-1.7467],
        [ 0.3405],
        [-0.5766],
        [ 1.4849],
        [ 0.9828],
        [-0.5527],
        [-1.2732],
        [-0.6007],
        [ 1.5537],
        [-0.0713],
        [ 0.0442],
        [-0.5992],
        [-0.4269],
        [-0.3135],
        [ 1.2982],
        [-0.7517]], device='cuda:0', requires_grad=True)
Parameter containing:
tensor([ 1.0732, -0.1505, -0.2502,  0.8316,  1.5188, -0.4151, -0.1245,  0.1900,
        -0.1961, -0.5356, -0.6462, -0.2152, -0.3987,  0.6659,  0.9365, -0.3791,
        -0.1930, -0.4840,  0.1964, -0.6832], device='cuda:0',
       requires_grad=True)
Parameter containing:
tensor([[ 3.6130e-01, -1.9921e-01,  3.6066e-02,  3.3548e-01,  3.3947e-01,
          6.8570e-02,  1.9454e-01, -1.4379e-02, -1.4283e-01,  1.3451e-01,
          3.2939e-02,  2.2855e-02, -6.5641e-02,  2.1689e-01,  1.1170e-01,
         -9.7901e-03, -1.2513e-01,  4.8520e-02, -1.2761e-01,  2.0919e-02],
        [-8.9221e-01, -1.4780e-01,  7.4627e-01, -1.2753e+00, -1.1121e+00,
          2.4015e-01, -1.9078e-01, -3.4960e-02,  4.4754e-01,  2.3385e-01,
```

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          1.5574e-01, -2.5064e-01, -1.6421e-01, -2.0571e-01, -2.1005e-01,
          2.3030e-02, -1.8649e-01, -1.3138e-01, -2.1300e-01,  1.1858e-01],
        [ 2.6585e-01, -1.0059e-01, -2.1921e-01,  3.7745e-01, -2.9998e-02,
          9.7056e-02,  3.0887e-02,  2.0462e-01, -6.7770e-02,  8.9547e-02,
          9.7436e-02,  1.1603e-01, -1.0849e-01,  5.7636e-02,  1.1554e-01,
         -1.9482e-01,  9.9974e-02,  5.7191e-02, -1.4751e-02, -2.9126e-02],
        [-3.7367e-03, -2.7401e-01,  1.0882e-02, -3.3449e-01, -2.0108e-01,
         -1.4589e-01, -1.3305e-01,  1.6256e-01,  1.6227e-01,  1.1682e-01,
          1.4027e-01, -1.9980e-01, -2.3191e-01,  4.4557e-02, -1.2850e-01,
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        [ 1.8461e-01,  2.9775e-02, -4.5512e-01,  6.1278e-01,  4.5119e-01,
         -2.7571e-01,  2.2362e-01, -9.5329e-02, -4.6886e-01,  1.3574e-01,
          1.9032e-01,  2.4689e-01, -1.8539e-01,  1.0320e-02, -5.4519e-02,
          1.9653e-01, -7.6328e-02, -1.3315e-01,  9.2000e-02,  8.2242e-03],
        [ 8.3177e-02,  4.6000e-02, -9.3681e-02, -2.3540e-02,  8.9513e-02,
          8.5560e-03, -1.5401e-02, -8.7370e-02, -2.4662e-01,  7.7306e-02,
         -4.7886e-02, -1.6128e-01,  1.5436e-01,  8.7513e-02, -1.3381e-02,
         -9.6504e-02, -1.4308e-02,  1.5939e-01, -8.4323e-02,  1.3144e-01],
        [-6.7927e-02,  1.2020e-01, -1.4932e-01, -8.0218e-02, -3.3234e-01,
          1.9101e-01,  6.2338e-02,  1.6332e-01,  4.4408e-02,  1.9255e-01,
          2.0099e-01, -6.5212e-02,  1.4532e-01,  9.7459e-02, -1.3206e-01,
          7.3287e-02,  1.8743e-02,  8.0692e-02,  5.2334e-02, -1.0898e-02],
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          1.3090e-01,  3.8950e-02,  2.6872e-01, -3.7675e-02,  1.4534e-01,
         -1.1151e-01,  8.9570e-03,  2.7254e-01,  2.1781e-01,  8.7402e-02],
        [-8.5664e-02, -1.4718e-01, -6.6535e-02,  6.4443e-02, -2.3874e-01,
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         -4.2154e-02, -1.1662e-02, -1.6867e-01,  2.2122e-01, -5.2697e-02,
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        [ 3.1496e-01,  1.0408e-01, -6.2356e-01,  2.9011e-01,  3.3610e-01,
         -9.8649e-02,  1.1377e-01, -3.7959e-03, -2.2678e-01,  1.5996e-01,
          9.2111e-02,  1.6639e-01, -3.6829e-01, -1.5290e-01, -8.4935e-02,
          1.8335e-01,  8.8447e-02,  2.2065e-01,  6.7349e-02,  6.2422e-02],
        [ 3.5911e-01,  1.3640e-01, -2.1084e-01,  8.1778e-01,  4.9770e-01,
         -3.7739e-01,  1.0771e-01,  2.2900e-01, -8.3617e-02,  5.3268e-02,
         -1.5390e-01,  7.5515e-02, -1.4467e-01,  1.0520e-01,  5.7693e-03,
          8.8995e-02,  2.0478e-01, -5.5217e-02,  9.0288e-02, -4.8792e-02],
        [ 1.8099e-02, -2.6099e-02, -1.0014e-02, -5.3521e-02, -7.4946e-02,
         -7.3790e-02, -1.0700e-01,  1.3879e-01, -2.2359e-02,  2.2389e-01,
          1.3175e-01,  7.6152e-02,  1.3601e-02,  7.4497e-02, -3.4582e-02,
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        [-9.8212e-02, -9.3969e-02, -1.4445e-01,  6.5060e-02,  8.8299e-02,
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         -6.8077e-02,  5.4781e-02,  1.4346e-01,  1.4222e-02,  1.5860e-01,
          4.6293e-02,  2.5710e-01, -7.4007e-02,  1.7647e-01, -4.1955e-02],
        [ 6.5126e-02, -2.6336e-01,  2.4063e-02,  5.3679e-03,  7.9032e-02,
         -2.1764e-01, -2.3693e-01, -1.6590e-01,  6.4803e-02, -6.3827e-03,
         -2.0530e-01, -4.9831e-02,  5.2765e-02, -1.4592e-01, -1.0889e-01,
          1.5625e-01, -1.0660e-01, -1.7077e-01, -3.2340e-02, -2.1794e-01],
        [-4.7177e-01, -4.1434e-01,  6.3974e-01, -5.4278e-01, -4.7130e-01,
          4.1289e-01, -6.1444e-01,  5.2429e-01,  8.2111e-01, -4.7909e-01,
         -2.8371e-01, -3.2772e-02,  5.8179e-01,  2.2024e-01,  3.3762e-02,
```

```
          -3.8996e-01, -3.0161e-01, -1.0386e-01,  5.4921e-01, -3.5927e-01],
         [-4.4575e-02, -8.5497e-02,  1.4974e-01, -2.5804e-01, -1.3975e-02,
           2.9733e-01, -3.8417e-02,  2.4856e-01,  3.3465e-01,  2.1298e-02,
          -1.0443e-01, -1.5073e-01,  1.9184e-01, -1.2611e-01, -1.9401e-01,
          -6.8432e-02,  1.8605e-01, -1.6349e-02, -3.5775e-03, -2.2284e-01],
         [ 9.5760e-02, -1.3660e-01, -1.8143e-01, -2.1604e-01, -8.7937e-02,
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          -7.8117e-02, -2.5838e-01,  4.3678e-02,  1.9656e-01,  2.7059e-01,
           1.0928e-01,  9.2289e-02, -1.0748e-02, -4.1834e-02,  2.2459e-02],
         [ 7.0787e-02, -8.3696e-02, -4.8704e-01,  4.9867e-01,  3.4868e-01,
          -2.4395e-01,  1.7598e-01, -1.6139e-01, -1.2791e-01,  1.7632e-01,
           2.1380e-02,  1.4118e-01, -2.4809e-01,  4.2652e-02,  4.3887e-02,
          -6.9587e-02,  1.7802e-01, -2.0472e-01, -9.5927e-02,  2.0479e-01],
         [-1.6495e-01,  3.1169e-02, -7.7071e-02, -1.2322e-01, -1.2262e-01,
           1.7484e-02, -1.4813e-01,  7.1304e-02,  1.4518e-01, -1.9087e-01,
           1.1642e-01,  4.8145e-02, -2.6846e-01,  3.6102e-02, -7.8649e-02,
           9.5705e-02, -2.3510e-01, -2.0734e-01, -1.3224e-01, -1.7169e-01],
         [ 2.7625e-01, -7.7177e-03,  5.6468e-03,  6.5989e-01,  1.0855e-01,
          -2.5052e-01,  1.4041e-01, -7.9625e-02, -1.9043e-01, -1.6108e-02,
          -1.8179e-01,  6.8590e-02, -2.1372e-01,  2.1710e-01,  1.7434e-01,
           2.2498e-01, -5.5995e-02, -1.6559e-01, -2.2510e-03,  5.7869e-02]],
       device='cuda:0', requires_grad=True)
Parameter containing:
tensor([ 0.2431, -0.2288, -0.0340, -0.0788, -0.1709,  0.1193,  0.1668, -0.0191,
         0.1849,  0.1799,  0.1452, -0.2203, -0.2330,  0.1861, -0.1753, -0.0546,
         0.0681, -0.1540,  0.0767,  0.2209], device='cuda:0',
       requires_grad=True)
Parameter containing:
tensor([[ 0.2483, -0.1401,  0.1503, -0.1454, -0.1483, -0.0008, -0.0864,  0.2891,
         -0.1605, -0.2067,  0.4727,  0.1055,  0.0531, -0.1880,  0.4061,  0.2361,
         -0.2068, -0.1137, -0.3767,  0.1321]], device='cuda:0',
       requires_grad=True)
Parameter containing:
tensor([-0.0847], device='cuda:0', requires_grad=True)
===========================================================================
======
Layer (type:depth-idx)                   Output Shape              Param #
===========================================================================
======
PINNModel                                [1, 1]                    --
├─ModuleList: 1-5                        --                        (recursive)
│    └─0.weight                                                    ├─20
│    └─0.bias                                                      ├─20
│    └─1.weight                                                    ├─400
│    └─1.bias                                                      ├─20
│    └─2.weight                                                    ├─20
│    └─2.bias                                                      └─1
│    └─Linear: 2-1                       [1, 20]                   40
│    │    └─weight                                                 ├─20
│    │    └─bias                                                   └─20
├─Tanh: 1-2                              [1, 20]                   --
├─ModuleList: 1-5                        --                        (recursive)
│    └─0.weight                                                    ├─20
```

```
│    └─0.bias                                        ├─20
│    └─1.weight                                      ├─400
│    └─1.bias                                        ├─20
│    └─2.weight                                      ├─20
│    └─2.bias                                        └─1
│    └─Linear: 2-2              [1, 20]             420
│    │    └─weight                                    ├─400
│    │    └─bias                                      └─20
├─Tanh: 1-4                    [1, 20]             --
├─ModuleList: 1-5              --                  (recursive)
│    └─0.weight                                      ├─20
│    └─0.bias                                        ├─20
│    └─1.weight                                      ├─400
│    └─1.bias                                        ├─20
│    └─2.weight                                      ├─20
│    └─2.bias                                        └─1
│    └─Linear: 2-3              [1, 1]              21
│    │    └─weight                                    ├─20
│    │    └─bias                                      └─1
==================================================================================
======
Total params: 481
Trainable params: 481
Non-trainable params: 0
Total mult-adds (Units.MEGABYTES): 0.00
==================================================================================
======
Input size (MB): 0.00
Forward/backward pass size (MB): 0.00
Params size (MB): 0.00
Estimated Total Size (MB): 0.00
==================================================================================
======

进程已结束，退出代码为 0
```